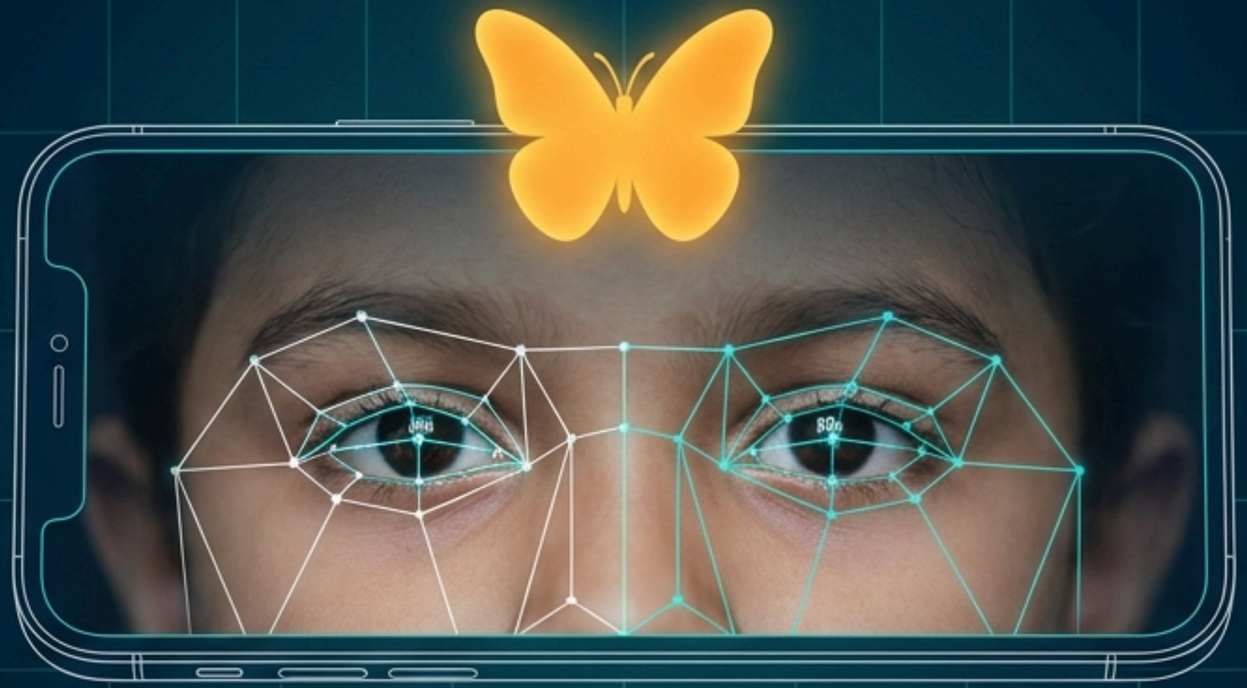
NETRA-SCREEN

Privacy-First Vision Pre-Screening for Children.

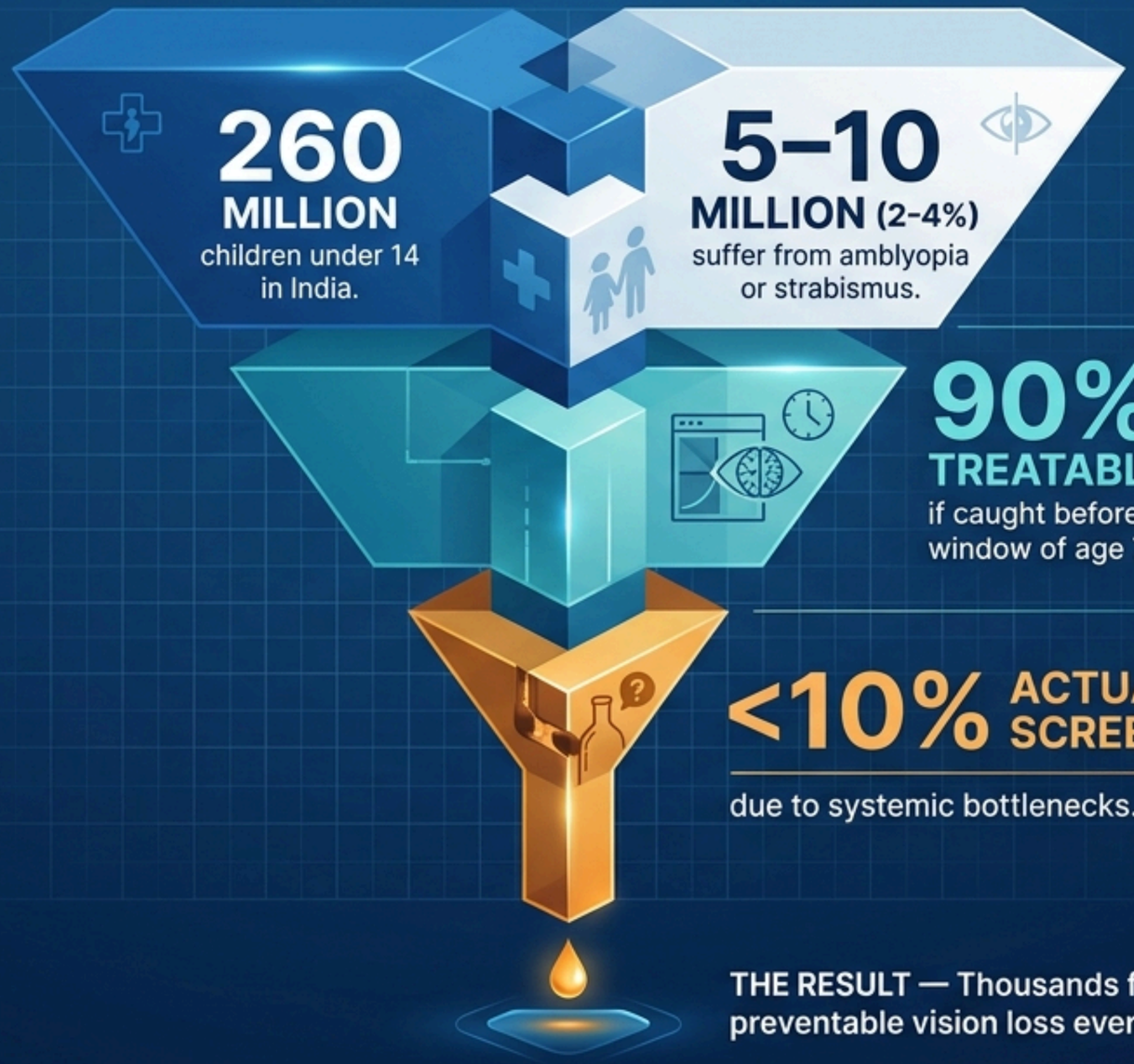
Turning any smartphone into an accessible, on-device AI screening tool.



Presented by
Team MindMesh | YIP 9.0

TKM Centenary
Public School, Kollam

The Silent Crisis in Pediatric Vision



260 MILLION
children under 14 in India.

5–10 MILLION (2-4%)
suffer from amblyopia or strabismus.

90% TREATABLE
if caught before the critical window of age 7–8 closes.

<10% ACTUALLY SCREENED
due to systemic bottlenecks.

THE RESULT — Thousands face permanent, preventable vision loss every year.

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The Accessibility Bottleneck

Traditional clinical screening equipment costs **₹1–3 Lakhs**. It is confined to urban clinics and requires specialized technicians.

Unavailable in **rural government schools** and **Anganwadi centers**. Current low-cost alternatives rely static letter charts (requiring literacy) or cloud processing (posing massive privacy risks).

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Democratizing Care via Smartphone AI

1. Engage (Child-Friendly)

The child follows animated targets (butterfly/fish/dot) directed across the screen. Zero literacy required.

2. Track (Smart Camera)

Harnesses the standard front camera and on-device AI to analyze binocular coordination in just 2 minutes. Automated quality gates instantly evaluate lighting, head pose, and frame quality.

3. Report (Instant & Bilingual)

Generates a local, fully offline research report (in Malayalam and English) to guide parental action. Zero extra hardware required.

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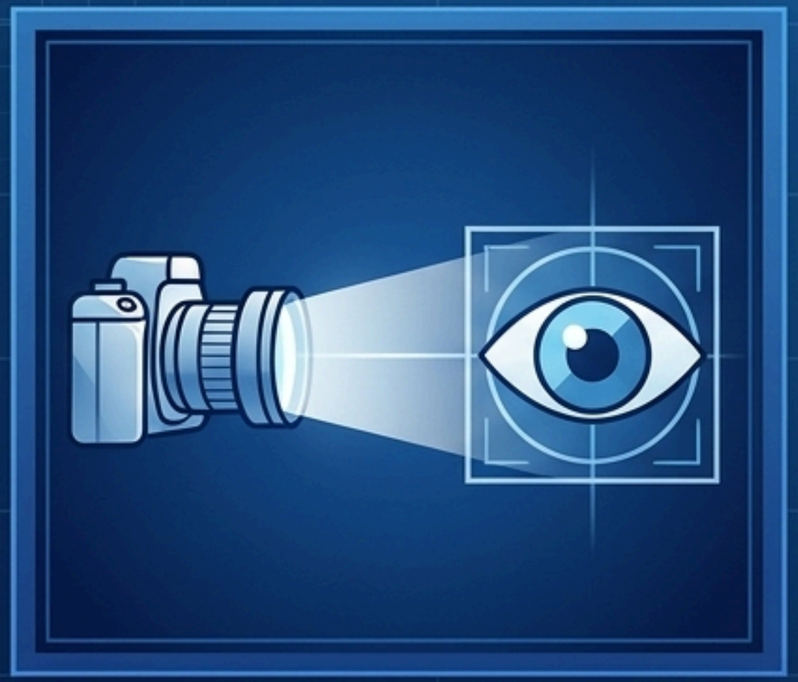
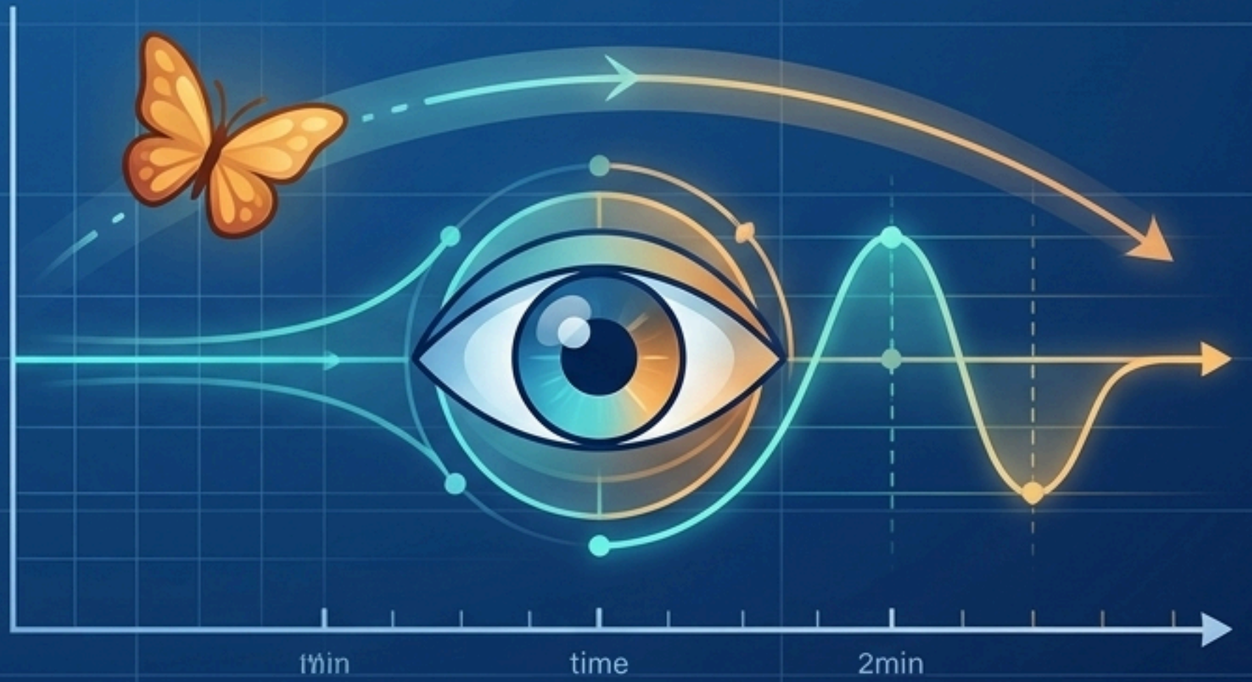
Dynamic Tracking vs. Static Checking





Traditional Static Tests:

Check visual acuity or light reflexes at a single, frozen moment in time. Cannot always capture functional coordination.

NETRA-SCREEN:

Analyzes functional movement. We study dynamic bilateral iris excursion symmetry during pursuit, and fixation stability during target holds. We watch how the eyes work together in real-time.

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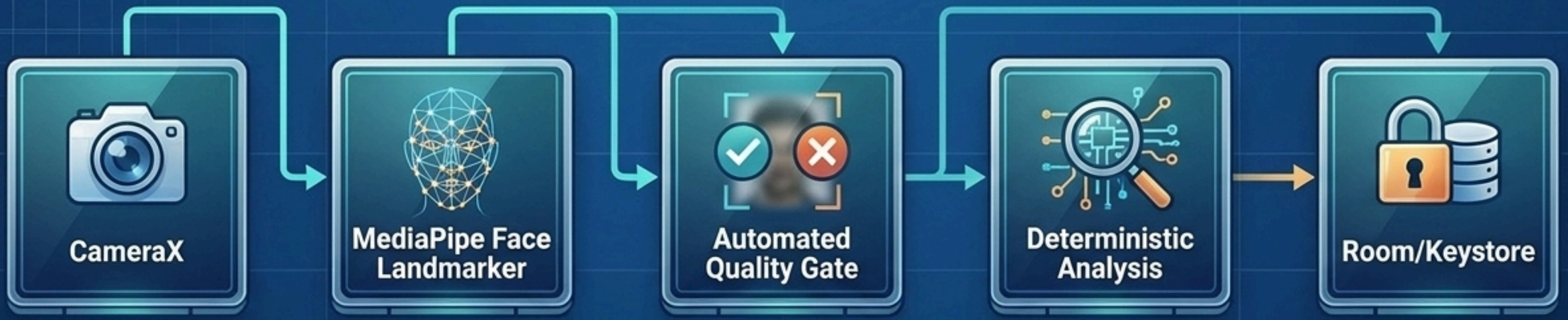
The Technology Engine



Data Layer:	Built natively on Android 8.0+ (Kotlin, Jetpack Compose). Covers 95% of active devices.
Inference Layer:	Powered by bundled **Google MediaPipe Face Landmarker** (mapping 478 precise facial points entirely on the CPU).
Quality Layer:	**Automated heuristics** instantly reject poor lighting, motion blur, and eyelid occlusion before accepting data.
Storage Layer:	AES-GCM encryption via Android Keystore.

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Shifting the Paradigm in Pre-Screening

	NETRA-SCREEN	Traditional Clinic	Standard Apps
Cost	✓ ₹0	⚠ ₹1–3 Lakhs	⚠ ₹0–₹500
Methodology	✓ Dynamic Binocular	✗ Static Exam	✗ Static Acuity/Letters
Accessibility	✓ 100% Offline Anywhere	✗ Urban Clinics Only	⚠ Requires Internet
Privacy	✓ On-Device Local	✓ In-Person	⚠ Cloud Storage Risks
Engagement	✓ Animated Games	✗ Intimidating	✗ Boring Tests

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Absolute Privacy by Design

- **Core Philosophy:** We are dealing with children's medical data. Privacy cannot be an afterthought.
- **100% On-Device:** No cloud inference, no internet permission required.
- **No Videos Retained:** Only numerical landmark measurements are processed.The raw camera frames are discarded instantly.
- **Cryptographic Security:** Summaries are secured using AES-GCM encryption in the app sandbox. Parents maintain total control to erase data instantly.

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Responsible Medical AI: Radical Honesty

What NETRA-SCREEN is NOT:

- It is not a diagnostic device.
- It does not measure true corneal flash reflections (Hirschberg reflex).
- It cannot confirm or exclude an eye disease.

What NETRA-SCREEN IS:

- It is a pre-screening triage tool using landmark-based research proxies.
- It identifies children who need an ophthalmologist.
- Clinical referral classification is explicitly disabled in v1.0 pending rigorous, independent medical validation. We do not make false claims.

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The Clinical Validation Roadmap (AAPOS Guidelines)

Current

Next 3 Months

Next 6-12 Months

18+ Months

Phase 1 - Technical Feasibility & Prototype
Complete. (YIP 9.0 Demo)

Phase 2 - Usability Assessment
Testing with 20-30 children for engagement metrics and workflow refinement.

Phase 3 - Prospective Clinical Validation
Partnering with ophthalmologists for a 100+ children study. Comparing app proxies against independent professional masked exams.

Phase 4 - Threshold Development
Setting safe referral limits and piloting real-world deployment in government schools.

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The Innovators Behind the Screen

Team MindMesh | YIP 9.0 Category 1

Malavika Basanth
(Team Leader)

Lavanya Vivek Pillai
(Member)

Devasree P C
(Member)

Proudly representing TKM Centenary Public School, Kollam.
Engineering passion for accessible, equitable healthcare innovation.

With Heartfelt Gratitude

Our Project Mentor

Aswathy P

Thank you for your guidance, encouragement and continued support throughout the NETRA-SCREEN project.

Our School

TKM Centenary Public School, Kollam

We sincerely thank our school for giving us the opportunity to explore, innovate and present our ideas through YIP 9.0.

Team MindMesh | NETRA-SCREEN | YIP 9.0

The Vision & The Ask

Zero Cost. Infinite Impact.

The Vision:

- Deploy NETRA-SCREEN via ASHA workers and school nurses using their existing personal smartphones. No new hardware, just scalable software.

The Impact:

- Reaching the 90% of children who currently fall through the cracks of the healthcare system.

The Ask:

- We are seeking YIP's support, mentorship, and clinical partnerships to launch our Phase 2 Usability Study. Let's ensure no child loses their vision simply because of where they live.

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